

Jun Yamamoto

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EDUCATION

Central European University, *Department of Network and Data Science* Sep. 2023 — Present
Doctor of Philosophy in Network Science GPA: 4.00/4.00
Supervisor: Prof. Dr. Márton Pósfai & Prof. Dr. János Kertész
Thesis: “Toward a general theory for physical networks dynamics” (tentative)

Queen Mary University of London, *School of Mathematical Sciences* Sep. 2022 — Sep. 2023
MSc in Mathematics Mark: Distinction
Supervisor: Prof. Dr. Ginestra Bianconi
Thesis: “Modelling higher-order network dynamics in the presence of triadic interactions”

ETH Zürich, *Department of Physics* Sep. 2019 — May 2020
Exchange Semesters Abroad (Terminated early due to the COVID-19 pandemics)

Hokkaido University, *Department of Applied Physics* Apr. 2017 — Mar. 2022
Bachelor of Engineering in Applied Science and Engineering GPA: 4.18/4.30
Supervisor: Prof. Dr. Kousuke Yakubo
Thesis: “Bifractality of fractal scale-free networks”

PUBLICATION

Journal paper

1. M. Niedostatek, A. Baptista, **J. Yamamoto**, J. Kurths, R. Sanchez Garcia, B. MacArthur, and G. Bianconi “Mining higher-order triadic interactions.” *Nature Communications* **16**, 11613 (2025). DOI: 10.1038/s41467-025-66577-z.
2. K. Yakubo, G. Shimojo, & **J. Yamamoto** “Random walks on bifractal networks.” *Physical Review E* **110**, 064318 (2024). DOI: 10.1103/PhysRevE.110.064318.
3. **J. Yamamoto** & K. Yakubo “Bifractality of fractal scale-free networks.” *Physical Review E* **108**, 024302 (2023). DOI: 10.1103/PhysRevE.108.024302.

PRESENTATION

ORAL

*Presenter is underlined.

1. **J. Yamamoto**, I. Bonamassa, and M. Pósfai “Localization-delocalization transition in physical networks.” The 29th International Conference on Statistical Physics (StatPhys29; Florence), July 14, 2025.
2. **J. Yamamoto**, I. Bonamassa, and M. Pósfai “Delocalization of Physical Laplacian Eigenvectors due to Degree-Volume Correlation.” The International School and Conference on Network Science (NetSci2025; Maastricht), June 4, 2025.
3. **J. Yamamoto**, I. Bonamassa, and M. Pósfai “Reentrant localization in physical Laplacian eigenvectors” DYNASNET workshop on graphs, networks, and brains, May 13, 2025 (Budapest, Hungary).
4. **J. Yamamoto**, I. Bonamassa, and M. Pósfai “Laplacian Spectra and Angular Constraints in Physical Networks” Annual DYNASNET Meeting, October 16, 2024 (Valtice, Czech).
5. **J. Yamamoto** and K. Yakubo “Bifractal property of stochastic scale-free networks,” The Physical Society of Japan 77th Annual Meeting, March 15, 2022 (Online).
6. **J. Yamamoto** and K. Yakubo “The multifractality of scale-free networks,” The Physical Society of Japan 76th Annual Meeting, September 20, 2021 (Online).

POSTER

1. **J. Yamamoto**, G. Shimojo, and K. Yakubo “Bifractal nature of fractal scale-free networks and its implications,” Network Science Seminar 2022 (Kyoto), August 23–25, 2022.
2. **J. Yamamoto** and K. Yakubo “Bifractality of scale-free networks,” Network Science Seminar 2021 (Kanawaza), December 11–12, 2021. *Poster Award*.

AWARD & SCHOLARSHIP

First-Year PhD Award, Central European University
Awarded for outstanding coursework and performance during the comprehensive exam. Vienna, Austria
Feb. 2025

Principal's Prize, Queen Mary University of London
Awarded for outstanding academic achievements during MSc studies. London, United Kingdom
Nov. 2023

ITO Scholarship Award for Japanese Students,
ITO Foundation for International Education Exchange Tokyo, Japan
Awarded ¥3,000,000 tuition fee & \$2,000/month stipend for Japanese students studying master's degree abroad. Sep. 2022 — Sep. 2023

Dean's Award for Academic Achievement, School of Engineering, Hokkaido University Sapporo, Japan
Awarded to students with outstanding academic achievements at the School of Engineering. Mar. 2022

Lane Memorial Award, Hokkaido University Sapporo, Japan
Awarded to the eight students with outstanding grades in English in the first and second years of undergraduate studies. Jul. 2019

Scholarship for Study Abroad under Agreement, Japan Student Service Organization Tokyo, Japan
Awarded ¥80,000/month for the duration of the exchange programme. Sep. 2019 — May 2020

Nitobe Award, Hokkaido University Sapporo, Japan
Awarded to the best student at each school by GPA of the first year. Jul. 2018

TEACHING

Central European University

- **Scientific Python**, Department of Network and Data Science Fall AY2025/26
- **Introduction to Linear Algebra**, Undergraduate Program Winter AY2024/25

Hokkaido University

- **Statistical Mechanics I**, School of Engineering Spring AY2022/23
- **Applied Mathematics II**, School of Engineering Spring AY2022/23
- **Computational Science**, Spring AY2022/23
Education and Research Centre for Mathematics and Data Sciences Fall AY2021/22
- **Introduction to Statistics** (As a part of Mathematics and Data Science/AI Seminar Series),
Education and Research Centre for Mathematics and Data Sciences Spring AY2021/2022

ACADEMIC SERVICE

Review Activity

- **Nature Communications** Since 2025.
- **Transactions of the Japanese Society for Artificial Intelligence** Since 2025.

Organization

- **Organizer, Physical Networks III Satellite** Dec. 2024 — Jun. 2025
The International School and Conference on Network Science (NetSci2025) Maastricht, Netherland
- **Organizer, Data Stories 11th Edition** Nov. 2024 — May 2025
Department of Network and Data Science, Central European University Vienna, Austria

OTHER EXPERIENCE

Fujitsu, DX Large Data Platform Business Group Tokyo, Japan
Data Scientist Jul 2022 — Sep 2022

- Analysed large-scale datasets of newspaper articles using natural language processing and network analysis.
- Quantified the trends of markets by employing the dynamic topic model and developed software that visualises the correlation between the trending topics/keywords and economic indices.
- Analysed the topology of collocation networks of keywords in the news articles and bipartite networks of keywords and newspaper articles.
- Developed a portfolio in which the collocation networks are visualised and used to recommend related news articles or keywords.

Japan Atomic Energy Agency
Summer Research Intern

Kashiwa, Japan
 Jul 2020 — Aug 2020

- Implement the self-learning hybrid Monte Carlo method to accelerate molecular dynamics.
- Simulate titanium dioxide using the self-learning hybrid Monte Carlo method to validate the method.

Okinawa Institute of Science and Technology
Research Intern

Onna, Japan
 Feb 2018 — Mar 2018

- Operate scanning electron microscope (SEM), environmental transmission electron microscope (ETEM), and focused ion beam (FIB).
- Construct a vacuum system and test cold field emitters in the system.
- Prepare samples of gold particles and carbon nanotubes for experiments.

WORKSHOP & SCHOOL

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| • IceLab Camp (Umeå, Sweden) | 23 — 26 Sep 2025 |
| • Workshop on Graphs, Networks, and Brains (Budapest, Hungary) | 12 — 13 May 2025 |
| • Winter Workshop on Complex Systems (Lambordy, Italy) | 26 — 31 Jan 2025 |
| • Annual DYNASNET Meeting 2024 (Valtice, Czech) | 14 — 18 Oct 2024 |

MEMBERSHIP

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| • American Physical Society | Since Jan 2026. |
| • Austrian Physical Society | Since Mar 2024. |
| • The Physical Society of Japan | Since Nov 2020. |

SKILL

- **Language:** Japanese (native); English (fluent); German (beginner)
- **Programming:** Python; C++; Julia; Fortran
- **Software:** Linux; Git; L^AT_EX; Mathematica